communication, aerodynamic design, propulsion systems, and mechanical control mechanisms.

At the heart of these early drones was the radio control (RC) system, which allowed human operators to guide an aircraft from the ground using wireless signals. This technology, still in its infancy at the time, relied on simple transmitters and receivers that could modulate a few basic flight commands such as turning, climbing, or descending. The lack of frequency stability and the limited range of early radio transmitters restricted operational distance, but this innovation represented the first step toward remote-controlled aviation.

In terms of aerodynamic design, first-generation drones primarily adopted fixed-wing configurations, similar to conventional airplanes. Their structure consisted of a main fuselage, wings, a tail assembly, and control surfaces, including rudders and elevators. The design focused on achieving basic lift and stability rather than agility or efficiency. The airframes were typically constructed from wood, canvas, and early lightweight metals, chosen for ease of fabrication and cost-effectiveness.

Propulsion technology for these drones was equally rudimentary. Most relied on small internal combustion or piston engines, which provided limited thrust suitable for short-range flights. Propeller-driven systems were manually started and offered minimal power regulation during operation. Despite their simplicity, these engines marked the beginning of motorized unmanned flight and introduced the principle of integrating propulsion with wireless control.

Mechanically, these drones operated through servo-based linkages and analog control systems that manipulated flight surfaces. Control was achieved through direct radio input, without any stabilization assistance or